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RESEARCH ARTICLE

YOLO-DefXpert: An Advanced Defect Detection on PCB Surfaces Using Improved YOLOv11 Algorithm

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ABSTRACT Surface defects in Printed Circuit Boards (PCBs), which arise during manufacturing, significantly impact product quality and directly influence equipment performance, stability and reliability. Accurately identifying small defects on PCB surfaces remains a considerable challenge, particularly under complex background conditions, due to the intricate and compact layout of the boards. This study introduces an improved PCB defect detection model, YOLO-DefXpert, using the YOLOv11 algorithm to address the low accuracy and efficiency challenges in detecting tiny-sized defects on PCBs. First, the standard backbone network of the YOLOv11 algorithm is replaced with the Swin Transformer to extract more robust features of defects, and the Convolutional Block Attention Module (CBAM) is added in the Patch Merging modules to alleviate feature leakage during the downsampling operation. Second, the standard convolutional operations are replaced with Deformable Convolutional Networksv2 (DCNv2) in the neck section to improve robustness in identifying multi-scale defects. Finally, an Additional Feature Fusion Layer (AFFL) is introduced in the neck to enhance the performance of the small defect identification. The effectiveness of the proposed YOLO-DefXpert is validated through experimental results obtained from publicly available PCB datasets. The proposed model achieves a mAP50 of 99.0% and a mAP95 of 60.6% in the HRIPCB benchmark dataset and 99.3% in mAP50 and 63.4% in mAP95 in the PCB dataset. Compared to the standard YOLOv11 model, the proposed YOLO-DefXpert attained an improvement of 9.3% and 13.2% in mAP50 and mAP95, an 11.25% increase in frames per second, and a 69.85MB decrease in model size. These findings highlight a notable enhancement in both accuracy and model efficiency in detecting tiny defects in the PCB board.

INDEX TERMS Convolutional neural network, deep learning, defect detection, printed circuit boards, small defect detection, YOLOv11.

I. INTRODUCTION

Industrial surface defect identification plays a critical role in various domains, including metal materials [1], power electronics [2], and aero engines [3]. Printed Circuit Boards (PCBs) are primary constituents of integrated circuits and are widely utilized across diverse manufacturing sectors, such as robotics, communication systems, automotive electronics, digital cameras, laptops and smartphones [4]. With the rapid progression of technology, the demand for higher-quality

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PCBs continues to grow. The automatic detection of surface defects in PCBs is particularly significant. However, surface defects during the PCB manufacturing process are unavoidable due to operator expertise variations, inconsistencies in production materials, and equipment maintenance [5]. Surface defects in PCBs not only compromise the reliability of the final products but can also result in the rejection of entire production batches, causing significant economic losses for manufacturers [6].

The commonly used PCB defect detection techniques are reference comparison, performance testing, manual inspection, and non-reference approaches [7]. Manual inspection relies heavily on technicians' personal assessment judgment to identify visible defects on the PCB with the naked eye. However, this method is inefficient and prone to oversight. Performance testing involves functional assessments of the PCB to detect defects, but this approach can result in physical harm to the board. The reference comparison technique, extensively adopted in the manufacturing sector, employs template-matching techniques. This method compares the PCB under examination to a standard template to identify defects. However, its effectiveness is significantly hindered by the need for precise spatial alignment, as misalignment can lead to a high false detection rate.

Over the past few years, Automatic Optical Inspection (AOI) has emerged as a leading method in industrial inspections [8]. Compared to traditional techniques, AOI offers improved detection accuracy. Nonetheless, the system's strict parameter settings and high sensitivity often lead to over-screening, necessitating additional manual secondary screening. In contrast to these traditional methods, defect identification techniques based on deep learning (DL) have demonstrated superior performance. They effectively address low detection accuracy and efficiency issues, making them a prominent research focus in recent years. With the rapid advancement of computer vision (CV) technologies, CV-based DL methods have gradually replaced conventional visual inspection approaches for PCB defect detection [9]. These methods excel due to their robust feature learning capabilities. Notably, CV-based DL techniques have emerged as established for PCB surface defect identification, eliminating the need for physical testing, which could potentially cause damage [10].

Object detection algorithms, in particular, have garnered significant attention for their ability to localize and recognize defects end-to-end manner simultaneously [11]. However, compared to general object detection tasks, accurately identifying PCB surface defects poses considerable challenges [12]: The sizes of defects vary depending on industrial scenarios and product types. This variation necessitates defect detection methods capable of adapting to multi-scale defect features. Complex backgrounds and subtle defects present significant challenges to defect detection. The intricate backgrounds in industrial images interfere with extracting and processing defect features. Furthermore, the unexpected occurrence of subtle defects, which are difficult to detect, adds another layer of complexity to the defect detection process. Defect features that deviate from the expected distribution disrupt normal feature interrelationships. Consequently, exploring and extracting high-order feature interrelationships are crucial for effective defect detection.

This paper introduces a new PCB defect detection model, YOLO-DefXpert, based on the YOLOv11 (You Only Look Once) algorithm [13] to tackle the above challenges. The primary objective of this model is to establish an efficient computer vision framework for detecting defects on PCB surfaces, maintaining commendable accuracy. The primary contributions of this paper are summarized below:

- To perform linear computation for multi-scale PCB images and to obtain hierarchical feature maps, the standard YOLOv11's backbone network CSPDarknet53 is replaced with a Swin Transformer network.
- The Convolutional Block Attention Module (CBAM) mechanism [14] is added to the patch merging and linear embedding blocks to mitigate feature leakage problems during downsampling operations.
- The standard convolution operation in the YOLO's neck section is replaced with Deformable Convolution (DCNv2) to strengthen the performance with lower-resolution feature maps.
- An Additional Feature Fusion Layer (AFFL) is added in the neck section for small defect detection. This layer improves the model's ability to capture shallow semantic features, thereby boosting the detection performance of tiny defects.

The structure of this paper is outlined as follows: Section II provides a concise review of existing methods for PCB defect identification. Section III presents a detailed explanation of the proposed model. In Section IV, the effectiveness of the proposed model, YOLO-DefXpert, is validated using a standard dataset and benchmarked against the performance of existing algorithms. Lastly, the article concludes with a summary of key findings.

II. RELATED WORKS

Automated inspection methods for PCBs can be distinguished into two types: Conventional machine learning-based and DL-based approaches. This paper begins with a review of the literature on traditional methods and then explores the more efficient and widely used DL methods. We then discuss the limitations identified in the current literature, concluding with an overview of the innovations in our approach to address these shortcomings.

A. TRADITIONAL APPROACHES FOR PCB DEFECT IDENTIFICATION

Before adopting machine vision in industrial quality inspection, the detection of PCB surface defects primarily relied on manual visual inspection. Raihan and Ce [15] proposed a detection method that integrates OpenCV with image

subtraction, enabling the rapid identification of defective PCB products. Similarly, Yang and Li [16] introduced the design of an automated PCB defect identification framework based on the AOI approach, capable of identifying open-circuit and short-circuit issues and line gap and scratch defects. Even today, many companies continue using this method to assess product quality. However, its low efficiency and high rate of missed detections have increasingly rendered it inadequate to meet modern industrial production demands.

Over the decade, machine learning-based (ML) algorithms have incorporated template-based approaches [17] for detecting PCB defects. These approaches utilize image-matching mechanisms to pinpoint defect features, which are examined using ML models. Hao et al. [18] proposed a method that integrates the neural network's capability to discover intricate defect structures with the optimization strengths of a genetic algorithm, specifically for real-time solder joint monitoring on PCBs. Compared to conventional techniques, this innovative method enhances solder joint deficiency identification accuracy and efficiency. Annaby et al. [19] introduced an enhanced normalized cross-correlation (NCC) technique for detecting defects in PCBs. Their approach improves upon the conventional NCC by integrating noise reduction and normalization techniques to achieve greater precision in defect detection. This method employs a sliding window technique to compare test images with the reference images pixel by pixel, utilizing the improved NCC to pinpoint divergences indicative of defects. Despite its effectiveness, this approach is computationally demanding because of the pixel-by-pixel comparison, which makes it less efficient for high-resolution images. Furthermore, it is less robust in uncontrolled environments, as it Battles with fluctuations in alignment and lighting.

Template-based methods face a significant obstacle: the requirement for flawless template images. Since various PCB inspection tasks necessitate unique template images, this dependency leads to considerable operational inefficiencies. Furthermore, most ML-based PCB monitoring models are restricted to defect classifying, the class of defect without pinpointing its position. These limitations stem from these algorithms primarily operating as classifiers, lacking the capability to provide positional information.

B. DL-BASED APPROACHES FOR PCB DEFECT IDENTIFICATION

DL techniques, especially YOLO, have recently gained significant traction in PCB defect detection. Li et al. [20] present a technique to tackle obstacles in manufacturing defect identification, explicitly focusing on minor defects that exhibit poor foreground-background segmentation. Their approach is an enhancement of the YOLO object detection framework. Several studies have focused on tailoring particular convolutional components or incorporating attention mechanisms into the benchmark architecture to enhance the detection accuracy of models. For instance, Du et al. [21] introduced a YOLO-MBBi model, which leverages an improved YOLOv5 DL framework to detect surface defects on PCBs. A PCB defect identification technique, PCB-YOLO, based on YOLOv5 [22], was introduced in [23], enabling the swift identification of surface defects on PCBs. Zhao et al. [24] integrated ShuffleNetV2 into YOLOv5, incorporating an attention mechanism and the concept of transfer learning to enhance defect detection performance.

Shen et al. [25] proposed PCBA-YOLO, an enhanced YOLOv5-based method that improves defect detection accuracy by incorporating large kernel convolution, an SIoU loss function, and modified neck structures. A new dataset, PCBA-DET, is introduced, providing annotated defect categories and locations. Wu et al. [26] enhanced the YOLOv5 detection algorithm by incorporating the Ghost module to decrease the spatial dimensions of the convolution layers. For feature extraction, the channel-focusing technique was introduced in the backbone layer. These modifications effectively addressed the challenges of manually detecting grotto murals, which often have low accuracy rates. Park et al. [27] examined the features of PCB images, categorizing fault identification using three different approaches. Their study employed ResNet50 and YOLOv7 to perform categorization and localization tasks, demonstrating the effectiveness of these models in identifying defects. Defects in printed circuit board assembly can significantly affect product quality and functionality.

Noroozi et al. [28] introduced a framework combining data preparation and detection modules to enhance defect detection, demonstrating its suitability for high-volume PCB production lines. Wu et al. [29] introduced a DL-based defect localization method, GSC YOLOv5, which combines a lightweight model with a double attention mechanism. The technique enhances YOLOv5 by incorporating Ghost Bottleneck and Ghost Conv layers to reduce network complexity. Additionally, the Squeeze-and-Excitation module and CBAM optimize detection performance. Jiang et al. [30] designed the DCR-YOLO network to enhance small-target defect detection on PCBs by balancing real-time speed and accuracy. It uses the DCR-backbone for feature extraction and the SDDT-FPN module for efficient feature fusion. The PCR module improves fusion across different scales, while the C5ECA module adapts feature weights for better small-target detection. Three YOLO-Heads predict defects at multiple scales, optimizing detection performance.

Yunpeng et al. [31] presented YOLOv8-TDD, an enhanced approach for detecting PCB defects by integrating Swin Transformers, Dynamic Snake Convolution (DySnakeConv), and Biformer into the YOLOv8 framework [32]. Swin Transformers [33] improves feature processing with shifted windows, while DySnakeConv adapts filter responses to varying textures. Biformer enhances contextual understanding through bidirectional processing, enabling more accurate defect detection [34]. This approach overcomes the limitations of traditional PCB inspection techniques. Kong et al. [35] proposed the SMC-YOLO defect detection network for

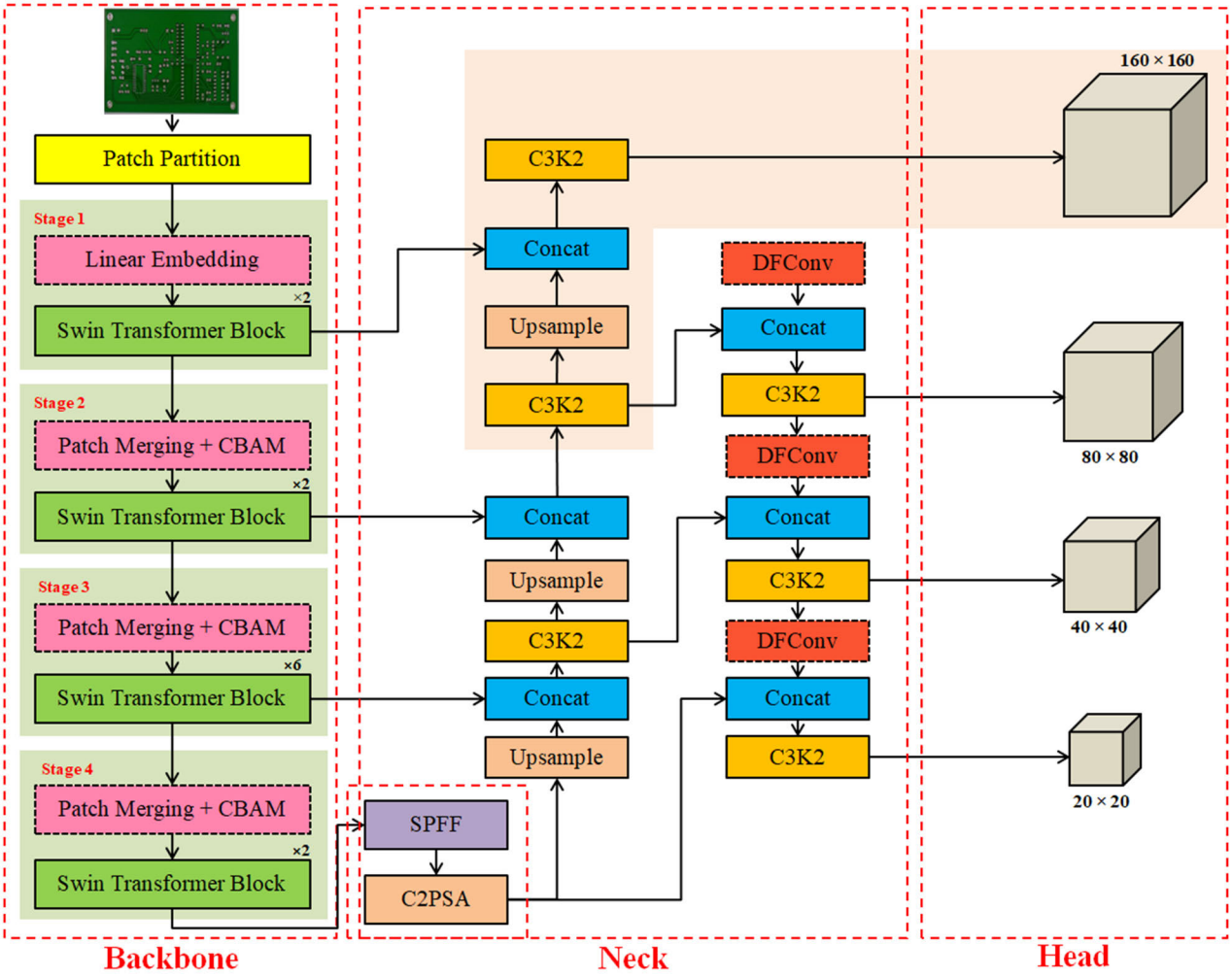


FIGURE 1. Pipeline architecture of the proposed PCB defect detection network, YOLO-DefXpert.

PCBs, addressing challenges like misidentification and low accuracy in detecting small defects. Enhancements to the SPPCSPC module, including additional convolutional layers and a feature channel, improve fine-grained feature capture. The multi-head self-attention mechanism and CARAFE operator enhance detection by focusing on critical features and restoring fine details. Additionally, integrating NWD and EIoU loss functions improves positioning accuracy and convergence efficiency.

III. PROPOSED MODEL

In 2024, Khanam and Hussain introduced the YOLOv11 algorithm, which achieved notable advancements in accuracy compared to other widely used object detection algorithms. An improved version of this model was developed to balance speed and accuracy, catering to the requirements of fast and efficient object detection tasks through its compact system design. As illustrated in Fig. 1, the proposed framework comprises of three primary sections: the backbone, the neck, and the head. The backbone handles feature extraction, the neck is responsible for feature fusion, and the head focuses on feature detection and output generation.

A. BACKBONE NETWORK USING SWIN TRANSFORMER

The standard CSPDarknet53 backbone is replaced with a hierarchical Swin Transformer architecture, which allows for more effective global and local feature extraction through shifted window-based self-attention. The Swin Transformer stages correspond to the hierarchical feature scales expected by YOLOv11's neck and detection heads. Specifically, the convolutional stem used in CSPDarknet53 is replaced by a patch partition layer that converts input images into non-overlapping patches. Subsequent stages of the Swin Transformer correspond to the downsampling stages of CSPDarknet53, producing feature maps at different scales. These multi-scale features are then passed to the custom neck and detection head, maintaining compatibility with YOLOv11's detection strategy. This integration maintains the spatial hierarchy required by YOLO's multi-scale detection

mechanism while leveraging the representational power of transformer-based attention modules.

The Transformer network was initially developed for transduction and sequence modeling tasks. Transformer is eminent for its self-attention mechanism of capturing long-range dependencies. It's a remarkable achievement in the Natural Language Processing domain directed researchers to explore its adaptability to computer vision problems, where it has recently exhibited potential success in vision problems. However, it demonstrated considerable hurdles in the visual domain, mainly scale, because the visual components can fluctuate widely in scale. In addition to the hurdle mentioned earlier, processing high-resolution images takes quadratic computational complexity.

On the other hand, Swin Transformer takes linear computational complexity for multi-scale image resolution and builds hierarchical feature maps. Hence, we incorporated Swin Transformer in our backbone network. We further improve the Swin Transformer by alleviating feature leakage during the patch downsampling process by adding the CBAM mechanism into the patch merging and linear embedding blocks. This allows the proposed model to distinguish multi-scale PCB defects.

The backbone network consists of a patch partition block and four different stages, 1 to 4. Stage 1 comprises of two sequential Swin Transformers and a linear embedding block. Stage 3 consists of 6 Swin Transformer blocks and a patch merging block. Stages 2 and 4 contain two consecutive Swin Transformer blocks and a patch merging block. The Patch Partition module takes an RGB image of size 640×640 as its input and divides the input image into 4×4 non-overlapping patches, resulting feature map of $160 \times 160 \times 48$.

The linear embedding layer in Stage 1 projects the raw 48-dimensional patch features into a new space with a higher embedding dimension (in this study, $C = 96$). This transformation allows the model to better capture semantic information by learning a richer patch representation. This embedding layer improves the feature representation of each patch by capturing interactions between pixel values in the patch, making the features more expressive for downstream tasks. Patch merging block in each stage merges 2×2 patches into one, reducing feature map size by $2\times$ while increasing depth. The stage 2, stage 3, and stage 4 produce feature maps of $80\times80\times192$, $40\times40\times384$, and $20\times20\times768$, respectively.

Swin Transformer blocks at each stage process features while preserving their spatial dimensions. The workflow of a Swin Transformer block is depicted in Fig. 2. Each block comprises of two types of windows: a regular window and a shifted window. Both windows incorporate a Multi-Layer Perceptron (MLP), Layer Normalization (LN) and residual connections.

Their primary difference is in their attention mechanism: the regular window uses Window Multi-head Self-Attention (W-MSA), while the shifted window uses Shifted Window Multi-head Self-Attention (SW-MSA). The W-MSA reduces computational complexity while preserving spatial

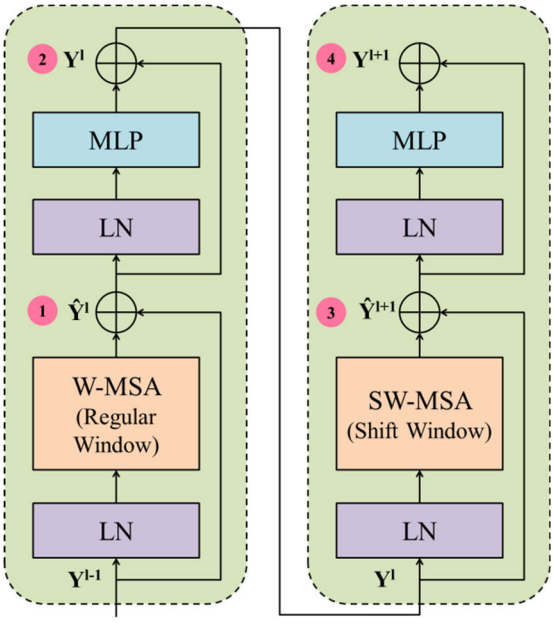


FIGURE 2. Architecture of the swin transformer blocks.

relationships and efficiently captures local context within non-overlapping windows. On the other hand, SW-MSA captures cross-window connections and helps the model efficiently compute local and global dependencies by performing self-attention within non-overlapping windows.

$$\hat{Y}^l = W - MSA(LN\left(Y^{l-1}\right) + Y^{l-1}) \quad (1)$$

$$Y^l = MLP(LN\left(\hat{Y}^l\right) + \hat{Y}^l) \quad (2)$$

$$\hat{Y}^{l+1} = SW - MSA(LN\left(Y^l\right) + Y^l) \quad (3)$$

$$Y^{l+1} = MLP(LN\left(\hat{Y}^{l+1}\right) + \hat{Y}^{l+1}) \quad (4)$$

where Y^{l-1} and Y^{l+1} represent the input and output of the Swin Transformer Block.

1) CONVOLUTIONAL BLOCK ATTENTION MODULE (CBAM)

The attention mechanism is a widely used technique for enabling adaptive focus on specific regions of an image within a neural network. The two popularly used attention mechanisms are spatial attention and channel attention. Channel attention allows the model to concentrate on category-specific information within an image by preserving the dimensionality of channels while reducing the spatial dimensions to a scalar. In contrast, spatial attention enhances the model's focus on the object's positional information within the image by maintaining the spatial dimensions and compressing the multi-channel information into a single channel.

This study employs the CBAM, which integrates channel and spatial attention to identify critical feature maps related to multi-scale PCB defects effectively. A schematic representation of CBAM is shown in Fig. 3. This block

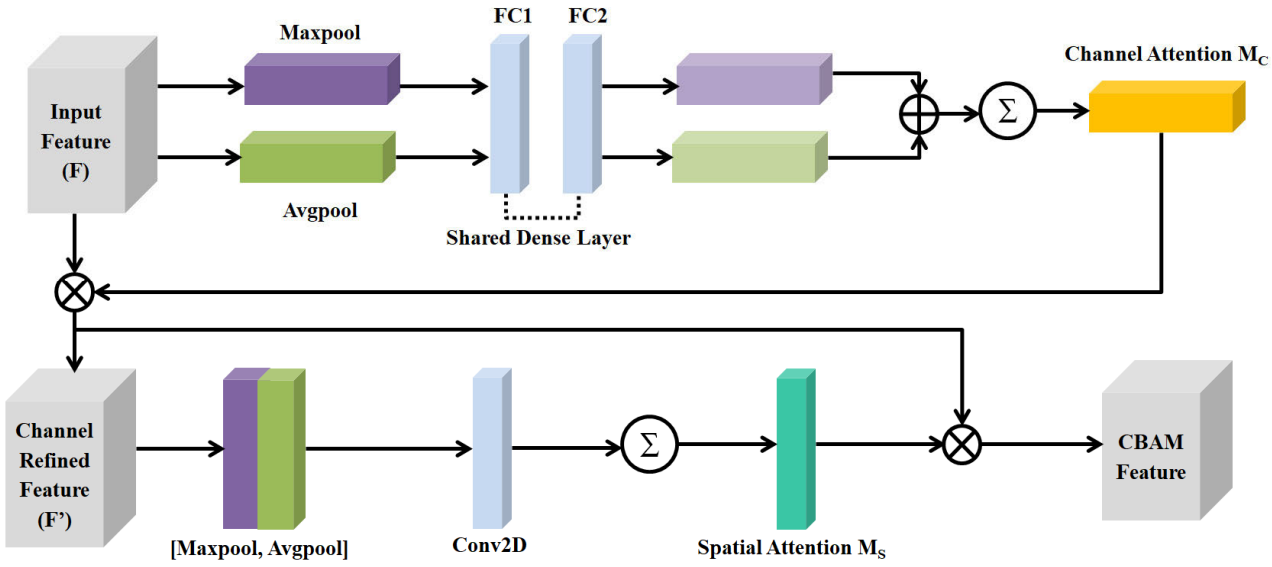


FIGURE 3. Schematic diagram of the CBAM network.

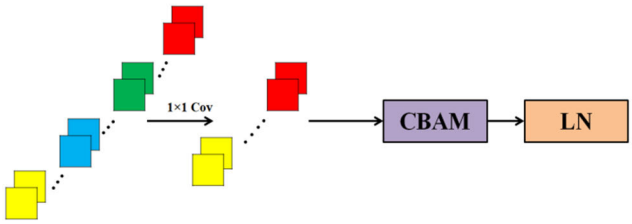


FIGURE 4. Flow diagram of the integration of CBAM with linear embedding block.

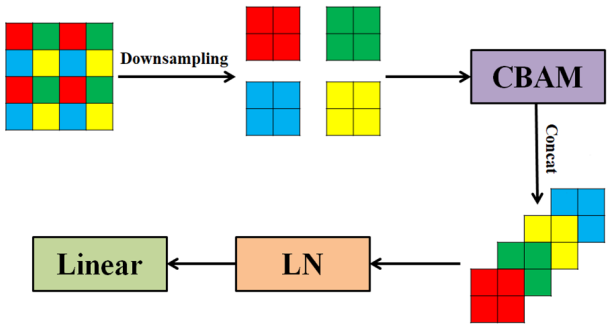


FIGURE 5. Flow diagram of the integration of CBAM with Patch merging block.

comprises two key mechanisms: the channel and spatial attention mechanisms. The integration of the CBAM into the linear embedding block is depicted in Fig. 4. The feature map's dimension, produced by the patch partition module, is adjusted to the channel dimension using a 1×1 convolution filter operation. The CBAM block is positioned ahead of the Layer Normalization (LN) layer.

Traditional downsampling techniques, such as convolution, max pooling, and average pooling, often operate within a local region, leading to inevitable feature leakage. On the other hand, patch merging chooses neighbourhoods of two adjacent pixels, reorganizes them into a set of patches (reducing their spatial dimensions by half), and combines the patches along the channel dimension (increasing the channel count by four). This process is finalized with a 1×1 convolution operation to refine the channel dimensions, ensuring that complete input data is preserved without feature leakage.

The integration of the CBAM module into the patch merging process is illustrated in Fig. 5. Within each channel, the neighbourhood areas of two points are restructured into a patch (effectively halving the spatial size). These reconstructed patches are then processed individually through the CBAM module. The aggregated feature maps from the CBAM module are subsequently combined along the channel dimension. The CBAM output is passed through an LN and fully connected linear layers to complete the process.

2) SPATIAL PYRAMID POOLING FAST (SPFF) BLOCK

The SPFF module in the improved YOLOv11 network pools features from different regions of an image at different scales. The pipeline structure of the SPFF block is shown in Fig. 6(a). This mechanism enhances the model's capability to locate targets of diverse sizes. The SPFF module achieves this by utilizing adaptive max-pooling with diverse kernel sizes, enabling the aggregation of multi-scale contextual information. By effectively combining details from different resolutions, the module allows the model to identify the smaller objects accurately.

3) CROSS-STAGE PARTIAL WITH SPATIAL ATTENTION (C2PSA)

The C2PSA mechanism leverages attention mechanisms to enhance the network's ability to concentrate on critical areas

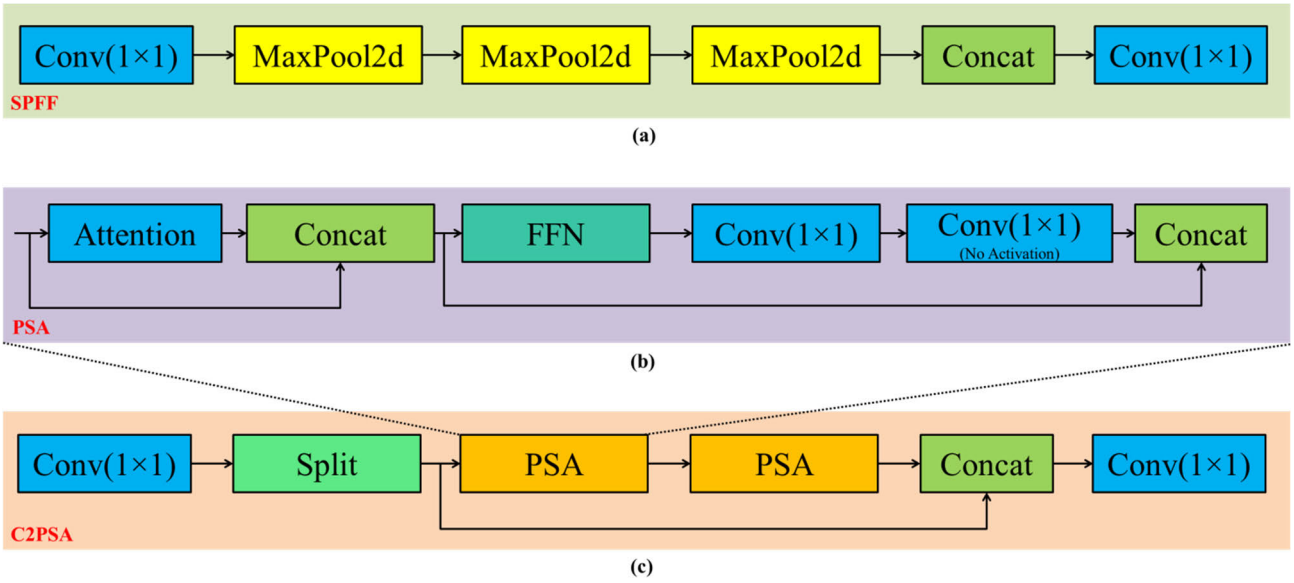


FIGURE 6. Pipeline architecture of: (a) Spatial pyramid pooling fast block. (b) Position-Sensitive attention block. (c) Cross-Stage partial with spatial attention (C2PSA).

within an image, particularly diminutive or obstructed targets, by reinforcing spatial context in the feature maps. The Position-Sensitive Attention (PSA) block is designed to apply PSA and feed-forward neural networks (FFNN) to input tensors, thereby strengthening feature extraction and computational efficiency. The process begins with the input layer being processed through an attention layer, followed by concatenation of the input and the attention layer's output. The block diagram of PSA and C2PSA blocks is depicted in Fig. 6(b) and Fig. 6(c).

The combined output is then passed through an FFNN, followed sequentially by a convolutional block with activation and another convolutional block without activation. Finally, the output of the second convolutional block is concatenated with the output of the initial concatenation layer, completing the process.

The C2PSA layer incorporates two PSA mechanisms operating on the feature map's separate branches. These branches are subsequently merged, like the C2F layered architecture. This design verifies that the network effectively captures spatial features while balancing computational efficiency and detection accuracy. By refining the network's capacity to concentrate on important regions selectively, the C2PSA block enables YOLOv11 to excel in earlier versions, such as YOLOv8, in environments where distinguishing delicate object features is crucial for achieving precise results.

B. NECK SECTION

The Neck layer is a crucial intermediary; it refines and aggregates the features extracted from different stages of the backbone and optimizes them for detection. The essential modules of the neck section are illustrated in Fig. 7.

1) C3K2 BLOCK

C3K2 blocks were employed to extract features at various levels of the backbone. The block diagram of C3K2 is depicted in Fig. 7a. The 3×3 kernels ensure computational efficiency while preserving the network's ability to acquire critical image features. This block enhances information flow within the model by dividing the feature map and employing multiple 3×3 convolutions, which are faster and reduce computational overhead. The C3K blocks (see Fig. 7b), in C3K2, feature a structure similar to the C2F block but do not perform feature map splitting. Instead, the input is directly processed through a convolutional block, followed by a chain of 'n' Bottleneck blocks with interconnections, and concludes with a final convolutional block. The flow diagram Conv and Bottleneck are blocks are illustrated in Fig. 7c and Fig. 7d. The C3K2 block builds upon the C3K block for efficient information processing. It includes two Conv blocks at the beginning and end, interspersed with a series of C3K blocks. Finally, the outputs from the last Conv block and the final C3K block are concatenated, culminating in another Conv block. This design balances speed and accuracy while leveraging the CSP structure effectively.

2) DEFORMABLE CONVOLUTIONAL NETWORKS v2 (DCNv2)

The convolution operation in the YOLO's neck section is mainly used for feature extraction. However, the traditional convolution operation has several limitations, mainly in multi-scale object detection. The traditional convolution operation has a fixed kernel size and fixed spatial connections, restricting the model's ability to capture complex and skewed objects, leading to poor performance in detection tasks for such objects.

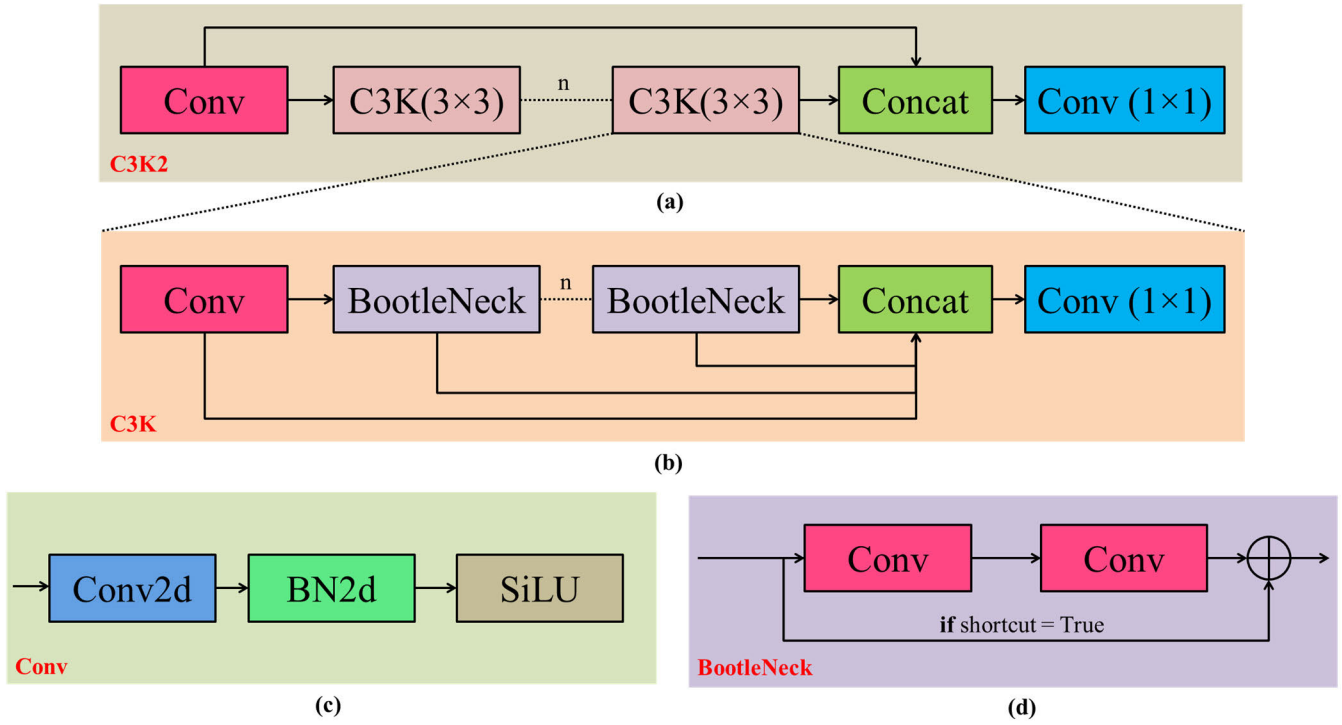


FIGURE 7. Pipeline architecture of: (a) Cross stage partial with kernel size 2 (C3K2). (b) C3K block. (c) Convolutional layer. (d) BootleNeck layer.

Deformable Convolution (DCNv2) [36] is an enhancement of classic convolution that enables the receptive field to adapt dynamically based on the input features. Replacing traditional convolutions with DFConv in the neck section of YOLO can provide several key advantages: (i) It detects arbitrary objects by allowing the receptive field to change based on the object's structure and location. (ii) It improves robustness in identifying multi-scale objects by allowing the model to focus on the most informative feature. (iii) It allows the network to obtain spatial features from fewer points in the receptive field, which helps the network to strengthen the performance with lower-resolution feature maps.

Considering these advantages, we have replaced standard convolution with DFConv in the neck section. The DCNv2 operation can be expressed as follows (see Eq. (5)):

$$R_{i,j} = \sum_{a,b} F_{i+a,j+b} \times W_{a,b,c} \tag{5}$$

where $R_{i,j}$ denote the resultant feature map, a and b are the offset values that discover the sampling position of the input feature map. The offset values are updated continuously during training and can further be represented as (see Eq. (6):

$$a = a_0 + \Delta a, \quad b = b_0 + \Delta b \tag{6}$$

where $(\Delta a, \Delta b)$ denotes the learned offset value, while (a_0, b_0) denote the receptive field's center position. These offset values are calculated based on the input feature map $(F_{i,j})$ and the output feature map $(R_{i,j})$, enabling the network to determine the optimal sampling position for each point.

The DCNv2 module adjusts offsets for perceiving input features. In extreme cases, this module completely ignores signals from specific locations by setting their feature amplitudes to zero, effectively reducing or eliminating their influence on the output. Furthermore, DCNv2 introduces modulated deformable convolution, which enhances the sampling grid's adaptability by incorporating a modulation factor into the weight tensor. This factor is scaled by the learned adjustment value, offering greater control over the sampling process. This modulation mechanism gives the network additional flexibility to adapt its spatial support regions (see Eq. (7).

$$R_{i,j} = \sum_{a,b} F_{i+a,j+b} \times W_{a,b,c} \times M_{a,b,c} \tag{7}$$

where, $M_{a,b,c}$ represents the modulation factor, calculated using the input feature map $(F_{i,j})$ and the output feature map $(R_{i,j})$. This modulation factor enables the network to adapt the weight tensor according to the targets' geometry and deformation, which is particularly beneficial for PCB defect localization.

C. HEAD SECTION

The quantity of detection layers in the proposed network has elevated from 3 to 4, producing features of various scales such as 20 × 20, 40 × 40, 80 × 80 and 160 × 160 help to detect larger, medium and smaller objects. In CNNs, shallow layers extract proximal features, capturing precise information such as edges and delicate image structures. In contrast, deeper

TABLE 1. Statistics of the HRIPCB dataset, including the type of defects, training, validation, and testing splits.

Defect Classes	HRIPCB			PCB		
	Train	Val	Test	Train	Val	Test
spurious_copper	805	101	101	94	11	11
missing_hole	795	99	99	93	11	11
mouse_bite	787	98	98	93	11	11
Short	786	98	98	94	11	11
Spur	781	98	98	93	11	11
open_circuit	771	96	96	94	11	11
Total	**4725**	**591**	**591**	**561**	**66**	**66**

layers focus on abstract semantic information after multiple convolution and pooling operations. However, this process often causes the features of tiny defects to be concealed or vanish. Since detailed and edge information is critical for accurate classification and localization, we introduced an additional output layer in the neck section designed explicitly for small target detection. This layer enhances the network's ability to capture shallow semantic features, thereby improving multi-scale defects' detection performance.

The newly constructed fourth output layer encounters two Upsample and Concat operations, followed by an additional Upsample operation to produce a 160 × 160 feature map. By performing Upsample and feature fusion at various scales, this feature map integrates more detailed semantics with refined spatial granularity, allowing for more accurate detection and discrimination of smaller objects.

IV. EXPERIMENT

A. DATASET

The efficiency of the YOLO-DefXpert was validated using the HRIPCB benchmark dataset [37] and the PCB dataset from Peking University [38]. The template board images are acquired using a 16-megapixel HD industrial camera furnished with a CMOS sensor, which can be operated via remote control or computer software. The original image size is 4608 × 3456 pixels. The training images are captured in proper orientation, whereas the testing images are captured in an arbitrary orientation.

The PCB dataset employed in this research was sourced from the publicly available collection released by Ding from the Intelligent Robot Open Laboratory at Peking University [39]. It comprises 693 images, each representing one of six distinct defect categories. On average, the original images measure 2777 × 2138 pixels in resolution. Table 1 describes the number of images and the number of images utilized for training, validation and testing by HRIPCB and PCB datasets with 6 different classes.

TABLE 2. Various training parameters used by the proposed model.

System Configuration	13th Gen Intel(R) Core(TM) i7-13700 2.10 GHz
Operating System	Windows 11 (64-bit)
GPU	Intel(R) UHD Graphics 770 (8 GB)
Initial Learning Rate	0.001
Optimizer	Adam
Batch Size	16
Train-Val-Test Ratio	80-10-10
Loss Functions	Box, Class, and Distribution Focal Loss
Activation Function	SiLU
Image Resolution	640×640
Number of epochs	200
Number of Classes	6
Weight Decay	0.0005

TABLE 3. Performance of the proposed modules in HRIPCB dataset, ✓ represent that the existing block used in the baseline model is replaced with the proposed module at the particular experiment. × represent the existing module used.

Exp.	Backbone	DCNV2	AFFL	mAP50 (%)	mAP95 (%)
(i)	×	×	×	89.7	47.4
(ii)	×	✓	×	90.2	47.8
(iii)	×	×	✓	91.6	48.3
(iv)	✓	×	×	92.0	49.4
(v)	×	✓	✓	93.8	51.6
(vi)	✓	×	✓	96.2	56.2
(vii)	✓	✓	×	97.8	58.4
(viii)	✓	✓	✓	**99.0**	**60.6**

B. IMPLEMENTATION DETAILS

The proposed model's training parameters and their corresponding values are presented in Table 2.

C. PERFORMANCE METRICS

Key metrics for evaluating the efficiency of the proposed framework in real-world defect identification are mean average precision (mAP), recall (R), and precision (P). These metrics collectively represent the model's effectiveness and accuracy in practical deployment settings. The calculations

TABLE 4. Performance comparison of the proposed model with existing popular object identification models and state-of-the-art PCB defect detection algorithms using the HRIPCB dataset.

	Models	mAP50 (%)						mAP50 (%)	mAP95 (%)	FPS	Model Size (MB)
		Missing Holes (%)	Short (%)	Mouse Bite (%)	Open Circuit (%)	Spurious Copper (%)	Spur (%)				
Popular Object Detection Models	YOLOv5 [22]	99.5	95.0	84.8	89.3	90.7	87.7	91.2	51.0	56.48	178.00
	YOLOv6 [39]	99.5	93.0	81.3	86.0	87.9	81.7	88.2	47.5	83.66	123.00
	YOLOv7 [40]	99.1	85.7	77.9	81.6	70.9	80.7	82.6	38.1	37.22	71.39
	YOLOv8 [36]	99.4	96.0	86.9	89.3	92.6	89.1	90.3	45.2	65.28	30.20
	YOLOv10 [41]	99.2	94.3	84.4	87.6	89.6	86.8	90.3	51.2	55.60	28.40
	YOLOv11 [13]	99.1	97.3	88.5	89.2	90.9	90.3	89.7	47.4	93.46	34.86
Existing PCB Detection Models	YOLO-HMC [42]	99.5	99.0	97.6	98.1	98.9	97.9	98.6	54.9	46.42	22.67
	GCC-YOLO [43]	99.5	99.2	98.3	98.6	96.1	97.3	97.1	48.5	68.00	14.50
	Light-PDD [44]	99.6	96.5	97.1	97.7	97.2	94.7	93.8	47.1	68.71	95.70
	Transformer-YOLO [45]	98.0	98.0	96.0	97.0	95.0	95.0	97.0	54.6	37.41	93.90
	PCB-YOLO [23]	99.5	98.5	90.7	**99.5**	96.7	90.9	96.0	55.2	46.72	92.30
	Liu et al. [46]	99.5	96.5	89.7	91.7	96.7	92.3	94.4	58.3	84.32	38.24
	YOLOv5_ES [47]	99.5	97.0	94.2	94.5	95.7	94.1	95.7	51.1	73.50	**7.78**
	SMC-YOLO [35]	99.8	99.7	94.8	95.1	96.3	**98.8**	97.4	52.7	**114.94**	25.14
Proposed Model		**99.7**	**99.8**	**98.8**	98.4	**99.1**	98.5	**99.0**	**60.6**	104.71	28.8

TABLE 5. Performance comparison of the proposed model with state-of-the-art PCB defect detection algorithms using the PCB dataset.

	Models	mAP50 (%)						mAP50 (%)	mAP95 (%)	FPS	Model Size (MB)
		Missing Holes (%)	Short (%)	Mouse Bite (%)	Open Circuit (%)	Spurious Copper (%)	Spur (%)				
Existing PCB Detection Models	YOLOv11 [13]	99.5	98.9	96.5	98.3	97.2	95.3	97.6	50.9	**111.10**	18.3
	Ji et al. [48]	98.9	99.2	98.7	98.9	99.7	97.6	98.8	56.3	72.50	37.8
	Li et al. [49]	98.5	96.2	91.8	91.5	90.5	89.5	92.9	46.7	160.00	8.4
	Xiao et al. [50]	98.9	97.6	97.8	98.3	98.6	98.3	98.3	51.1	128.00	22.0
	Wang et al. [51]	99.5	98.0	93.5	95.7	91.6	98.2	96.1	51.6	56.80	**5.2**
	Guo et al. [52]	99.5	99.0	97.0	90.2	93.0	91.6	95.1	48.2	48.60	36.6
	YOLO-WWBi [53]	99.5	98.4	98.8	97.7	94.9	90.2	96.6	50.5	90.20	30.1
	YOLO-SSW [54]	99.5	99.3	98.1	98.5	97.9	97.0	98.4	57.3	114.80	23.7
Proposed Model		**99.8**	**99.6**	**99.5**	**99.1**	**99.0**	**98.7**	**99.3**	**63.4**	104.71	28.8

for P, R, and mAP are as follows: Eqs. (8) to (11).

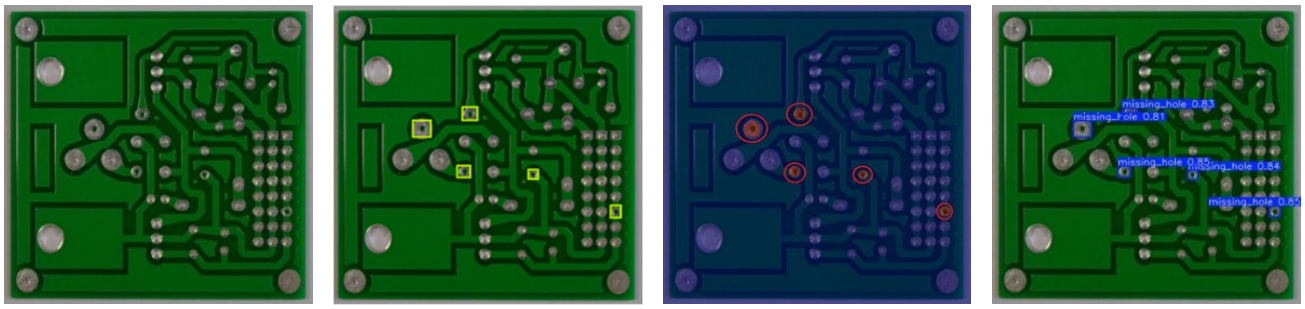
$$P = \frac{\text{TruePositive(TP)}}{\text{TruePositive (TP)} + \text{FalsePositive(FP)}} \tag{8}$$

$$R = \frac{\text{TruePositive(TP)}}{\text{TruePositive (TP)} + \text{FalseNegative(FN)}} \tag{9}$$

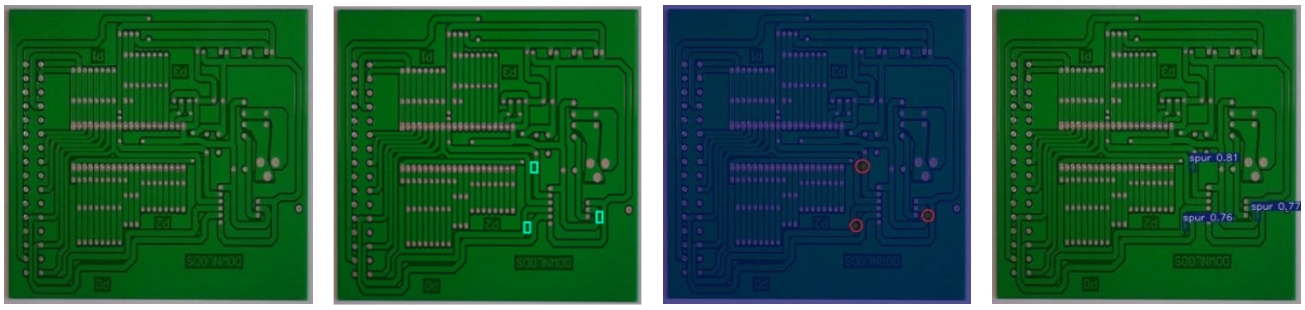
$$AP = \int_0^1 P(R)dR \tag{10}$$

$$mAP = \frac{\sum_{n=0}^{N} AP_n}{N} \tag{11}$$

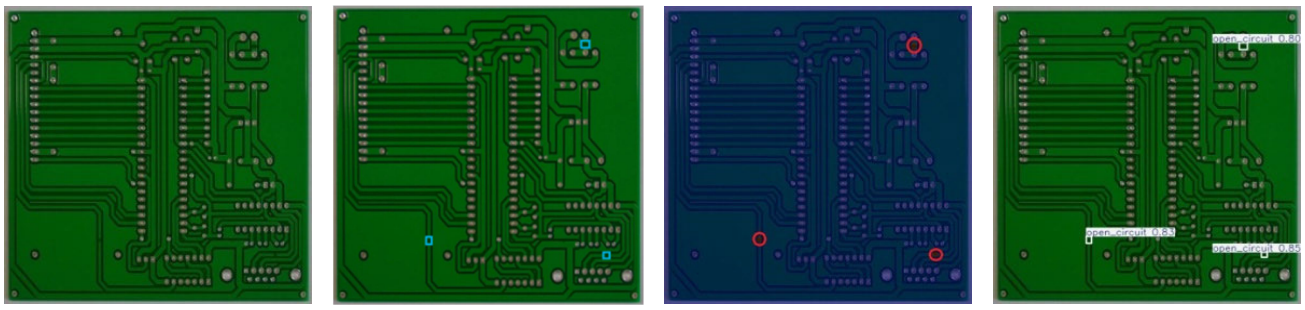
where, TP represents instances where the model correctly identifies an actual PCB defect. FN refers to cases where



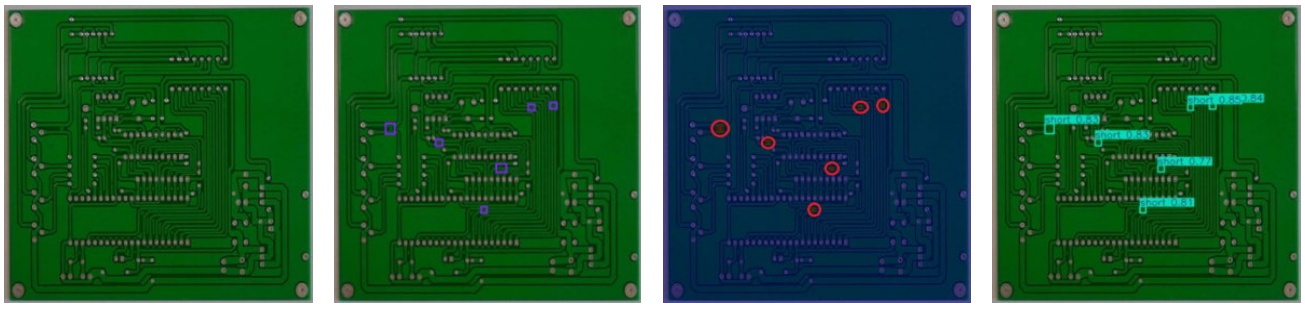
Missing Holes



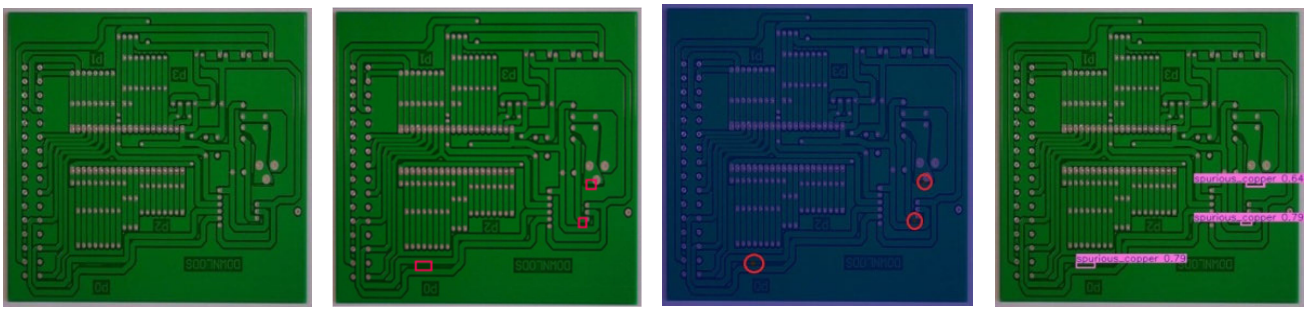
Spur



Open Circuit

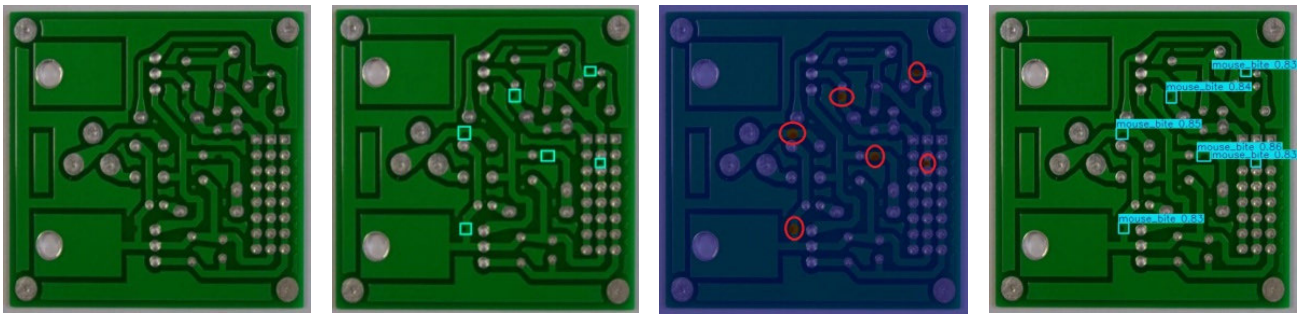


Short



Spurious Copper

FIGURE 8. Visualization of sample results HRIPCB dataset. From left to right, 1st column - input image, 2nd column - ground truth, 3rd column - attention map, and 4th column - detection result with bounding box, confidence score and class label.



Mouse Bite

FIGURE 8. *(Continued.)* **Visualization of sample results HRIPCB dataset. From left to right, 1st column - input image, 2nd column - ground truth, 3rd column - attention map, and 4th column - detection result with bounding box, confidence score and class label.**

the model does not identify a PCB defect. FP occurs when the model incorrectly predicts a PCB defect for a non-defective instance. N represents the total number of classes. AP_n denotes the average precision for class 'n'.

D. ABLATION STUDY

Detecting defects in PCBs demands accurate identification, precise localization, and exceptional real-time performance. The ablation study was conducted during the model schema modification to analyze the significance of the proposed modules, such as the Swin Transformer-based backbone network, DCNV2 and AFFL. Their performance is presented in Table 3.

The Exp. (i) represents the baseline model, i.e. original YOLOv11 in Exp. (ii), the baseline model's standard convolutional operation in the neck section is replaced with DCNV2. Similarly, in Exp. (iii), an additional detection head is included in the head section to identify small defects. In Exp. (iv), the original YOLOv11's backbone CSPDarknet53 was replaced with a Swin Transformer network. In Exp. (i) to (iii), we have conducted the study by replacing the baseline modules one by one with proposed modules at each iteration.

Whereas in Exps. (v) to (vii), we experimented with replacing two modules simultaneously. In Exp. (v), we have replaced the standard convolutional operation with DCNV2 operation and added AFFL in the neck section, we have used the baseline's backbone network. In Exp. (vi), the backbone network is replaced, and AFFL is included. In Exp. (vii), the same detection head of the baseline model with three feature maps is used, whereas the backbone and standard convolutional operation were replaced. In Exp. (viii) all the proposed modules were replaced with the existing modules in the baseline model, which achieved an average improvement of 9.3% in mAP50 and 13.2% in mAP95.

Integrating modules such as the Swin Transformer, CBAM, DCNv2, and AFFL aims to enhance feature extraction and attention mechanisms to boost performance on complex PCB datasets. While these components contribute significantly to the accuracy improvements (as shown in Table 4), they also increase the number of parameters, thus affecting training and inference time.

To address this, we performed a comparative analysis between our proposed model and lighter variants (e.g., baseline YOLOv5 and YOLOv8), showing that although our model incurs additional complexity, the gain in performance particularly in terms of mAP95 and robustness on challenging samples justifies the added computational cost for applications where accuracy is prioritized. The proposed model was primarily trained and evaluated using a combination of CPU and integrated GPU resources, which introduced certain limitations in terms of training speed.

E. PERFORMANCE COMPARISON WITH STATE-OF-THE-ART MODELS

This study compares its performance with several advanced PCB surface defect detection networks, including YOLOv5 [22], YOLOv6 [39], YOLOv7 [40], YOLOv8 [36], YOLOv10 [41] and YOLOv11 [13], to validate the reliability of the proposed network in identifying PCB surface defects. Additionally, experiments were conducted using seven improved algorithms [23], [35], [42], [43], [44], [45], [46], and [47] that are recognized for their effectiveness in detecting minor defects within the PCB dataset. The outcomes of these thirteen models on the PCB dataset are summarized in Table 4.

The data analysis in Table 4 leads to the following conclusions: the proposed model exhibits exceptional performance in mAP50 and mAP95, significantly surpassing other advanced PCB surface defect detection networks. Specifically, the proposed model achieves impressive mAP50 and mAP95 values of 99.0% and 60.6%. Compared to YOLOv5 [22], YOLOv6 [39], YOLOv7 [40], YOLOv8 [36], YOLOv10 [41] and YOLOv11 [13], it demonstrates mAP50 improvements of 7.8%, 10.8%, 16.4%, 8.7%, 8.7%, and 9.3%, respectively, underscoring its ability to accurately localize tiny defects and address the challenge of low sensitivity to minor defects.

The experimental results in Table 4 demonstrate that the proposed model outperforms other small target detection

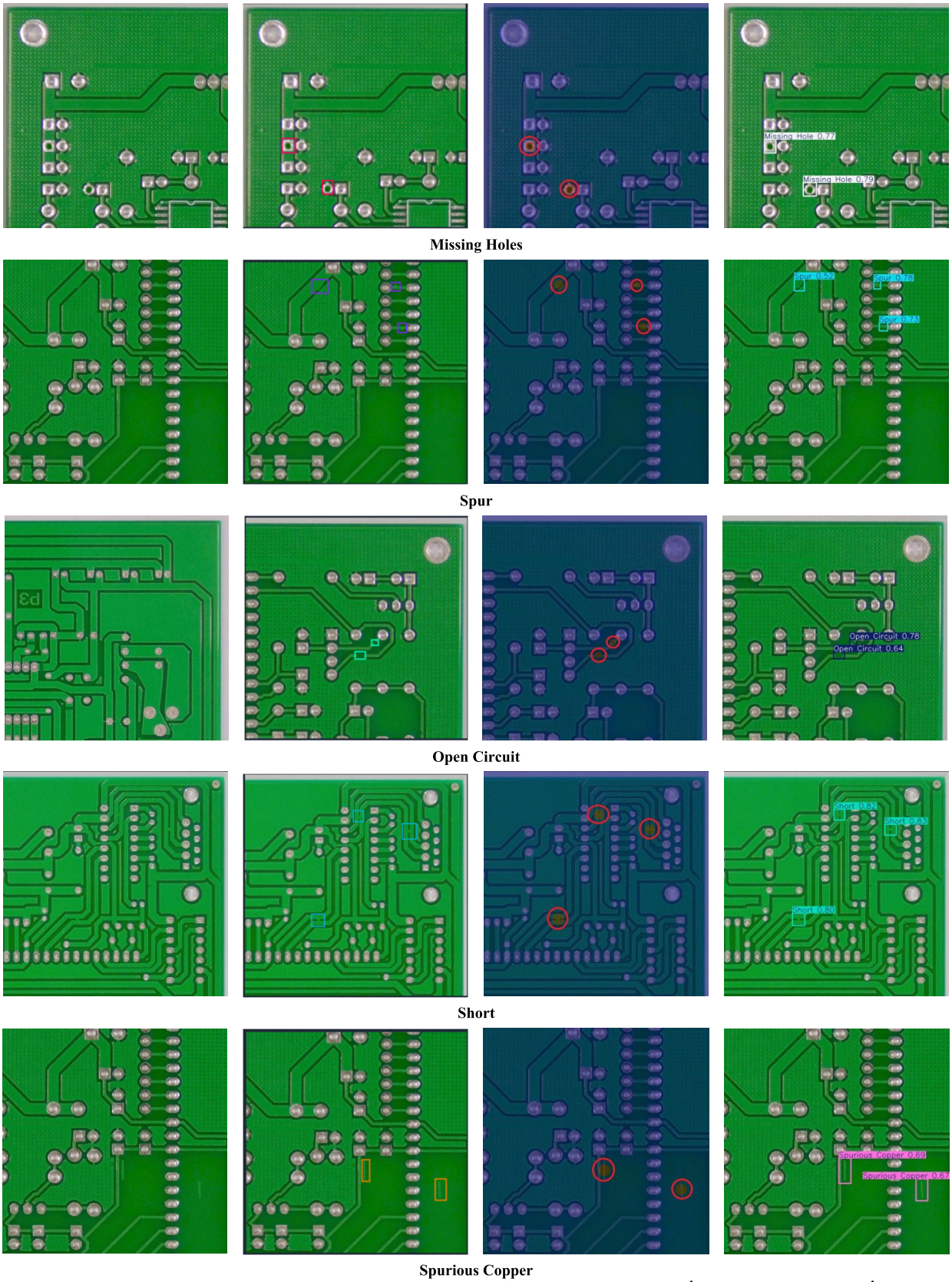
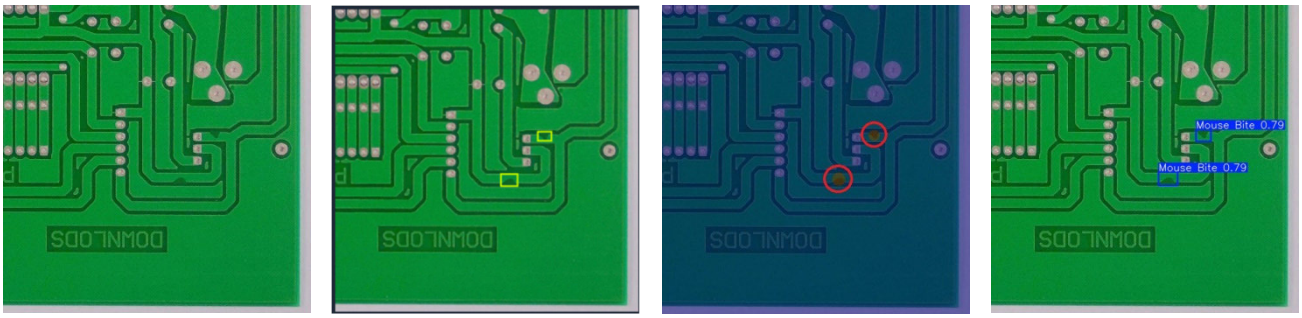


FIGURE 9. Visualization of sample results PCB dataset. From left to right, 1st column - input image, 2nd column - ground truth, 3rd column - attention map, and 4th column - detection result with bounding box, confidence score and class label.



Mouse Bite

FIGURE 9. *(Continued.)* **Visualization of sample results PCB dataset. From left to right, 1st column - input image, 2nd column - ground truth, 3rd column - attention map, and 4th column - detection result with bounding box, confidence score and class label.**

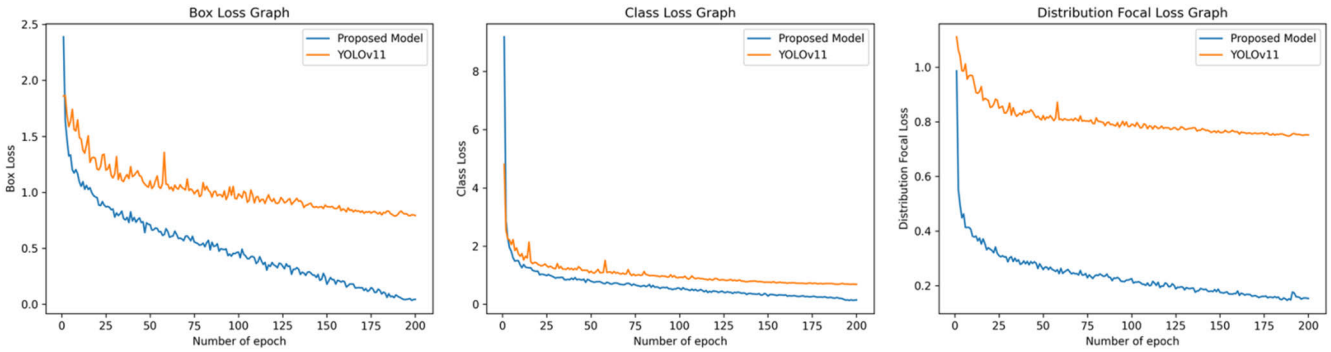


FIGURE 10. **Illustration of loss graphs: left-to-right, Box loss, Class loss and Distribution Focal loss.**

models, achieving the highest mAP50 and mAP95 values of 99.3% and 60.6%, showing superior performance in PCB defect detection, indicating that the proposed model can effectively locate many small targets. Furthermore, it delivers superior detection accuracy across nearly all defect types, except for the open circuit and spur, showcasing its exceptional performance in tiny PCB defect identification.

The empirical comparisons reveal that the proposed model significantly enhances detection accuracy for different defect classes compared to mainstream networks. Specifically, the mAP50 values for missing holes, short, mouse bite, and spurious copper are 99.7%, 99.8%, 98.8% and 99.1%, respectively, surpassing those of other algorithms. The proposed model operates at 104.71 FPS with a size of 28.75 MB, significantly outperforming almost all the models, slightly less than SMC-YOLO [35]; this makes the proposed model highly adaptable for industrial applications. Although the proposed model size is marginally higher than YOLOv10 [41], GCC-YOLO [43], YOLOv5_ES [47], and SMC-YOLO [35], it remains lower than many existing object models and improved PCB defect detection models [23], [44], [45], and [46] indicating that the proposed improvements do not significantly increase model complexity, making it well-suited for the industrial demands of PCB defect detection.

We also compared the performance of the proposed model with current state-of-the-art approaches on the PCB dataset, as shown in Table 5. Compared to our baseline model conventional YOLOv11 algorithm [13], the proposed improved YOLOv11 model attains an improvement of +12.5 in the mAP95 metric. Ji et al. [48] and Li et al. [49] achieved the state-of-the-art performance, but these techniques find it challenging in detecting tiny defects. The proposed model detects the maximum number of minor defects in all six classes and outperforms these techniques with a significant margin of +7.1 and +16.7 in mAP95 metric. Existing methods, such as Guo et al. [52] and YOLO-WWBi [53] performed on average in the Spur class, which is considered as one of the primary defects in PCB defect detection process, 91.6% and 90.2% of mAP50, that is less than ∼7.8% less than the proposed model's score. Similarly the proposed model outperofrms popular PCB defect detection techniques such as Xiao et al. [50] and Wang et al. [51] in all the six classes with the greater margins, such as +3.7% in mAP50 and +12.1% in mAP95, showcasing the significance of the proposed modules in the model. The proposed model outperforms most object detection models and their improved versions in all six classes in the HRIPCB and PCB datasets.

F. VISUALIZATION OF RESULTS

To better illustrate the detection performance of the proposed model, visualization results are provided in Figs. 8 and 9. The left-to-right columns display the original images, ground truth, corresponding attention graph, and detection results. The defects in HRIPCB are consistently small in size, and the background is highly complex, which significantly increases

the difficulty of detection. The proposed model effectively extracts visual features of these minute defects, achieving superior performance across all six defect classes. The proposed model effectively identifies various types of tiny defects, achieving an mAP50 average of 99% in HRIPCB.

During the training of the proposed defect identification network, the aggregate loss function is computed as the sum of box loss, class loss, and distribution focal loss. A lower total loss function value indicates better overall model performance, while a higher value reflects poorer performance. As depicted in Fig. 9, YOLO-DefXpert demonstrates a decreased loss function convergence value than the original YOLOv11 [13] algorithm during the training process. These results in higher detection precision, improved localization accuracy, and enhanced confidence and classification performance.

In summary, the visualization of detection results highlights our model's exceptional capabilities and strong generalization across various industrial applications. These include object tracking in security systems, facial recognition, obstacle detection for autonomous vehicles, traffic sign detection, and defect detection in industrial production lines. Compared to existing methods, the proposed model excels in three key aspects: effectively extracting and analyzing visual features of tiny defects, demonstrating robustness in handling complex backgrounds, and ensuring generalization across diverse industrial products.

V. CONCLUSION

This paper introduces a novel PCB defect identification model based on the YOLOv11 algorithm, YOLO-DefXpert. The proposed model utilizes Swin Transformers as its backbone, which takes linear computational complexity to extract distinctive features of small PCB defects by producing hierarchical feature maps. The CBAM mechanism is added to the embedding blocks of the backbone network to alleviate feature leakage problems during downsampling operations. In the neck section, the standard convolution operations are replaced with deformable convolutions to strengthen the performance with lower-resolution feature maps. An additional output layer in the neck section enhances the network's ability by capturing shallow semantic information to locate small PCB defects. Extensive experimental results exhibit the exceptional performance of YOLO-DefXpert compared to the popular object detection models and state-of-the-art PCB defect identification networks, excelling in both precision and efficiency. YOLO-DefXpert improved +9.7% in mAP50 and +13.2% in mAP95 over the baseline model YOLOv11. Additionally, ablation studies verify the significance of each proposed module, enabling the framework to achieve precise detection.

In future work, we aim to apply the YOLO-DefXpert model to a wider range of real-world industrial target identification applications to evaluate its practicality and generalization capabilities further. Furthermore, striking an optimal balance between detection accuracy and model size remains an important area for continued exploration.

ACKNOWLEDGMENT

Code and pre-trained weights will be made publicly available soon after internal approval and after the completion of ongoing expansions of this work.

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